SEFU HMIS — Database Architecture & Data Dictionary

Kenya Digital Health Agency (DHA) Certification Compliance Document

Category: System Documentation — Section D: Database Architecture & Data Dictionary

System Classification: Health Information System (HIS) & Electronic Medical Record (EMR)

Database Engine: MariaDB 10.4+ / MySQL 8.0+ (InnoDB Storage Engine, `utf8mb4_unicode_ci`)

■ 1. Database Overview & Design Principles

The **SEFU HMIS** relational database is designed according to Third Normal Form (3NF) principles, balancing clinical transactional integrity (ACID compliance) with high-speed query indexing for hospital wards.

1.1 Technical Design Highlights

- **Character Encoding:** `utf8mb4` with `utf8mb4_unicode_ci` collation, supporting multilingual clinical documentation.
- **Transactional Integrity:** InnoDB engine enforces foreign key constraints and transactional isolation (`$this->db->trans_start()` and `$this->db->trans_complete()`).
- **Audit Trails:** Master audit tables (`logs`, `userlog`, `staff_mfa_otps`) capture user identifiers, timestamps, IP addresses, user agents, and action types.
- **Index Optimization:** Primary keys (`INT AUTO_INCREMENT`), composite indexes on date/patient queries, and indexed foreign keys ensure sub-second retrieval over large patient cohorts.

■■ 2. Entity-Relationship Architecture (Core Schema)

```
erDiagram
PATIENTS ||--o{ OPD_DETAILS : "has outpatient visits"
PATIENTS ||--o{ IPD_DETAILS : "has inpatient admissions"
PATIENTS ||--o{ PHYSIOTHERAPY : "receives physical therapy"
PATIENTS ||--o{ PEDIATRIC : "has child health encounters"
PATIENTS ||--o{ GYNAECOLOGY : "has obstetric encounters"
PATIENTS ||--o{ TRANSACTIONS : "incurs payments & bills"
PATIENTS ||--o{ APPOINTMENTS : "schedules consultations"

STAFF ||--o{ STAFF_ROLES : "assigned roles"
ROLES ||--o{ STAFF_ROLES : "defines privileges"
ROLES ||--o{ ROLES_PERMISSIONS : "configures CRUD"
STAFF ||--o{ LOGS : "generates audit events"
STAFF ||--o{ STAFF_MFA_OTPS : "authenticates via 2FA"

OPD_DETAILS ||--o{ PHARMACY_BILLS : "dispenses medications"
OPD_DETAILS ||--o{ PATHOLOGY_TESTS : "orders lab tests"
OPD_DETAILS ||--o{ RADIOLOGY_TESTS : "orders imaging"

IPD_DETAILS ||--|| BEDS : "allotted bed"
```

■ 3. Comprehensive Data Dictionary (Core Health Tables)

3.1 Table: `patients` (Master Patient Index)

Stores core patient demographics, identification, and medical record numbers.

Column Name	Data Type	Nullable	Key	Default	Description

`id`	`INT(11)`	NO	PRI	*Auto*	Unique Master Patient Index (MPI) Identifier.
`patient_name`	`VARCHAR(150)`	NO	MUL	NULL	Full legal name of patient.
`dob`	`DATE`	YES		NULL	Date of birth.
`age`	`INT(11)`	YES		NULL	Patient age in years.
`month`	`INT(11)`	YES		NULL	Additional age in months (for infants/pediatrics).
`day`	`INT(11)`	YES		NULL	Additional age in days (for neonates).
`gender`	`VARCHAR(20)`	NO		NULL	Patient gender (Male, Female, Other).
`marital_status`	`VARCHAR(50)`	YES		NULL	Marital status.
`blood_group`	`VARCHAR(10)`	YES		NULL	Blood group (A+, B+, O+, AB+, etc.).
`mobileno`	`VARCHAR(30)`	YES	MUL	NULL	Primary contact phone number.
`email`	`VARCHAR(150)`	YES		NULL	Patient contact email address.
`address`	`TEXT`	YES		NULL	Physical residential address.
`guardian_name`	`VARCHAR(150)`	YES		NULL	Next of kin or parent/guardian name.
`identification_number`	`VARCHAR(100)`	YES	MUL	NULL	Kenya National ID / Alien ID / Passport Number.
`is_active`	`VARCHAR(10)`	NO		'yes'	Record status ('yes' active, 'no' disabled).
`created_at`	`TIMESTAMP`	NO		CURRENT	Record creation timestamp.

3.2 Table: `physiotherapy` (Physical Therapy Specialty Module)

Stores physical rehabilitation consultations, pain scoring, and therapy sessions.

Column Name	Data Type	Nullable	Key	Default	Description
`id`	`INT(11)`	NO	PRI	*Auto*	Unique Physiotherapy encounter ID.
`bill_no`	`VARCHAR(50)`	NO	MUL	NULL	System-generated billing voucher (`PHYB#`).
`patient_id`	`INT(11)`	NO	MUL	NULL	Foreign key referencing `patients.id`.
`physiotherapist_id`	`INT(11)`	YES	MUL	NULL	Foreign key referencing `staff.id` (Specialist).

`session_date`	`DATETIME`	NO		NULL	Date and time therapy was administered.
`diagnosis`	`TEXT`	YES		NULL	Clinical diagnosis and presentation.
`pain_location`	`VARCHAR(255)`	YES		NULL	Anatomical location of pain/injury.
`pain_scale`	`INT(11)`	YES		0	Visual Analog Pain Scale rating (0 to 10).
`treatment_plan`	`TEXT`	YES		NULL	Rehabilitation protocol and modalities applied.
`duration_minutes`	`INT(11)`	YES		30	Duration of therapy session in minutes.
`apply_charge`	`DECIMAL(10,2)`	NO		0.00	Net professional service fee charged.
`generated_by`	`INT(11)`	NO		NULL	Staff ID who created the record.
`created_at`	`TIMESTAMP`	NO		CURRENT	Clinical record creation timestamp.

3.3 Table: `pediatric` (Pediatric Specialty Module)

Stores specialized child health records, growth parameters, and developmental consultations.

Column Name	Data Type	Nullable	Key	Default	Description
`id`	`INT(11)`	NO	PRI	*Auto*	Unique Pediatric encounter ID.
`bill_no`	`VARCHAR(50)`	NO	MUL	NULL	System-generated billing voucher (`PEDB#`).
`patient_id`	`INT(11)`	NO	MUL	NULL	Foreign key referencing `patients.id`.
`pediatrician_id`	`INT(11)`	YES	MUL	NULL	Foreign key referencing `staff.id` (Pediatrician).
`session_date`	`DATETIME`	NO		NULL	Consultation date and time.
`birth_weight`	`VARCHAR(50)`	YES		NULL	Neonatal birth weight (kg/g).
`current_weight`	`VARCHAR(50)`	YES		NULL	Weight at current presentation (kg).
`height`	`VARCHAR(50)`	YES		NULL	Stature/length at presentation (cm).
`temperature`	`VARCHAR(50)`	YES		NULL	Body temperature (°C).
`diagnosis`	`TEXT`	YES		NULL	Pediatric clinical diagnosis.

`apply_charge`	`DECIMAL(10,2)`	NO		0.00	Net consultation fee charged.
`generated_by`	`INT(11)`	NO		NULL	Staff ID who generated the record.
`created_at`	`TIMESTAMP`	NO		CURRENT	Record creation timestamp.

3.4 Table: `gynaecology` (Gynaecology & Obstetrics Specialty Module)

Stores women's health consultations, obstetric histories, and antenatal tracking.

Column Name	Data Type	Nullable	Key	Default	Description
`id`	`INT(11)`	NO	PRI	*Auto*	Unique Gynaecology encounter ID.
`bill_no`	`VARCHAR(50)`	NO	MUL	NULL	System-generated billing voucher (`GYNB#`).
`patient_id`	`INT(11)`	NO	MUL	NULL	Foreign key referencing `patients.id`.
`gynaecologist_id`	`INT(11)`	YES	MUL	NULL	Foreign key referencing `staff.id` (Specialist).
`session_date`	`DATETIME`	NO		NULL	Consultation date and time.
`lmp`	`DATE`	YES		NULL	Date of Last Menstrual Period.
`edd`	`DATE`	YES		NULL	Calculated Expected Date of Delivery.
`gravida`	`VARCHAR(20)`	YES		NULL	Gravida count (total pregnancies).
`para`	`VARCHAR(20)`	YES		NULL	Para count (viable births).
`abortion`	`VARCHAR(20)`	YES		NULL	Abortion / pregnancy loss count.
`clinical_notes`	`TEXT`	YES		NULL	Gynaecological examination notes.
`apply_charge`	`DECIMAL(10,2)`	NO		0.00	Net professional service fee charged.
`created_at`	`TIMESTAMP`	NO		CURRENT	Clinical record creation timestamp.

3.5 Table: `logs` (Clinical & Administrative Audit Trails)

Captures all database record modifications across the entire HMIS for regulatory auditability.

Column Name	Data Type	Nullable	Key	Default	Description
`id`	`INT(11)`	NO	PRI	*Auto*	Unique audit record sequence number.

`message`	`TEXT`	YES		NULL	Human-readable description of change.
`record_id`	`INT(11)`	NO		NULL	Primary key ID of the affected entity.
`user_id`	`INT(11)`	YES	MUL	NULL	ID of user who executed the change.
`user_type`	`VARCHAR(20)`	NO		'staff'	Actor classification ('staff', 'patient', 'system').
`user_role`	`VARCHAR(50)`	NO		'Staff'	Actor role at time of change ('Doctor', 'HR', etc.).
`user_name`	`VARCHAR(150)`	YES		NULL	Resolved display name at the moment of action.
`action`	`VARCHAR(50)`	NO		NULL	SQL Action type ('Insert', 'Update', 'Delete').
`ip_address`	`VARCHAR(50)`	NO		NULL	Origin IPv4 or IPv6 client address.
`platform`	`VARCHAR(50)`	NO		NULL	Operating system platform of client.
`agent`	`VARCHAR(50)`	NO		NULL	Browser / User-Agent signature.
`time`	`TIMESTAMP`	NO		CURRENT	Exact timestamp of execution.
`created_at`	`DATE`	YES		NULL	Calendar date of action.

3.6 Table: `staff_mfa_otps` (Multi-Factor Authentication Engine)

Manages 2FA Email OTP tokens, expiration, and brute-force lockout states.

Column Name	Data Type	Nullable	Key	Default	Description
`id`	`INT(11)`	NO	PRI	*Auto*	Unique token sequence number.
`staff_id`	`INT(11)`	NO	MUL	NULL	Foreign key referencing `staff.id`.
`otp_code`	`VARCHAR(10)`	NO	MUL	NULL	Cryptographic 6-digit numeric token.
`email`	`VARCHAR(150)`	NO		NULL	Destination email address.
`expires_at`	`DATETIME`	NO		NULL	Token expiration timestamp (10 minutes).

`is_used`	`TINYINT(1)`	NO		0	State (`0=active`, `1=used`, `2=superseded`, `3=expired`, `4=locked`).
`attempts`	`INT(11)`	NO		0	Failed verification attempt counter (Lockout at 5).
`ip_address`	`VARCHAR(50)`	YES		NULL	Client IP requesting the token.
`created_at`	`TIMESTAMP`	NO		CURRENT	Token issuance timestamp.

Certified by:

Lead Database Architect & CTO

SEFU Health Systems — Kenya Digital Health Agency (DHA) Compliance Sign-Off